MIL-D-2222F
29 March 1985
SUPERSEDING
MIL-D-2222E
18 June 1969

MILITARY SPECIFICATION

DOUGH TROUGH: BAKER'S (FOR MOBILE BAKERY M-534)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of dough trough with body handles and cover.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet
QQ-S-571	-	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy
QQ-S-635	-	Steel Plate, Carbon
RR-C-271	-	Chains and Attachment, Welded, Weldless
TT-E-529	-	Enamel, Alkyd, Semi-Gloss

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
WW-T-700/3	-	Tube, Aluminum Alloy, Drawn, Seamless, 2024
MMM-A-1617	-	Adhesive, Rubber Base, General Purpose

MILITARY

MIL-B-2316	-	Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery Of
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum Alloys
MIL-R-5674	-	Rivets, Structural, Aluminum Alloy, Titanium Columbium Alloy, General Specification For
MIL-R-7885	-	Rivets Blind, Structural, Pull-Stem and Chemically Expanded

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-601	-	Rubber, Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MS 35792	-	Handle, Bail, Chest, Recessed

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

5-11-299	-	Dough Trough, Bakers' (For Mobile Bakery M-534); Illustration and Assembly
5-11-300	-	Details
5-11-301	-	Side, End and Bottom Panels
5-11-302	-	Cover Assembly and Details
5-11-303	-	Cover and Tie Bar Details

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 29 - Steel Bars, Carbon and Alloy, Hot-Wrought and Cold Finished, General Requirements For
A 167 - Chromium Nickel Steel, Stainless and Heat Resisting, Plate, Sheet and Strip
D 1565 - Cellular Materials, Flexible, Vinyl Chloride, Polymers and Copolymers (Open Cell Foam)
D 1621 - Plastics, Rigid Cellular, Compressive Properties of
D 2341 - Rigid Urethane Foam, Specification for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

* 3.2 Materials (see 6.3) and components. Materials and components requirements shall be in accordance with this document and subsidiary documents cited on the applicable drawings.

3.2.1 Steel, carbon.

3.2.1.1 Bar. Steel bar shall be grade 1010-1020 conforming to ASTM A 29.

3.2.1.2 Plate. Steel plate shall be SAE 1009 or 1020 conforming to QQ-S-635.

3.2.2 Stainless steel. Stainless steel sheet and strip shall conform to type 302 or 304 of ASTM A 167.

3.2.3 Aluminum alloy.

3.2.3.1 Sheet, 5052. Aluminum alloy sheet shall conform to 5052-H32 of QQ-A-250/8.

3.2.3.2 Sheet, 2024. Aluminum alloy sheet shall conform to 2024-T3 of QQ-A-250/4.

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3.2.3.3 Tube, 2024. Aluminum alloy tube shall conform to 2024-T3, type I of WW-T-700/3.

3.2.4 Hardware. All hardware shall be of the design, type, and material as shown on applicable drawing.

3.2.5 Chain. Steel chain shall conform to type I, grade C, class 5, style 2, hot dipped galvanized finish of RR-C-271.

3.2.6 Handles, recessed. Recessed handles shall conform to MS 35792-21 except countersunk steel screws are not required.

3.2.7 Rivets.

3.2.7.1 Solid aluminum rivets. Solid aluminum, 1/4 inch diameter, countersunk head, rivets shall conform to 2117-T4 of MIL-R-5674.

3.2.7.2 Blind rivets. Blind rivets shall conform to type II, class 1 of MIL-R-7885.

3.2.8 Solder. Solder shall conform to composition Sn 96 of QQ-S-571.

3.2.9 Insulation. The insulation shall be either rigid urethane foam conforming to properties of ASTM D 2341 or shall be expanded, closed-cell, rubber hardboard conforming to the requirements listed below when tested as specified in 4.4.1.1.

Water absorption, percent maximum	10
Density, pounds per cubic foot (pcf)	8 ± 1
Compressive strength, pounds per square inch (psi) minimum	40

3.2.10 Adhesive. Adhesive used for cementing the insulating board to the corrosion-resisting steel and aluminum sheets shall conform to type II of MMM-A-1672.

3.2.11 Primer. Zinc chromate primer shall conform to TT-P-1757.

3.2.12 Enamel. Enamel shall conform to TT-E-529. Color shall be Olive Drab No. 24087 conforming to FED-STD-595.

3.3 Construction. The construction of the dough trough shall conform to the requirements herein and the design, dimensions, and material classifications as shown on drawings listed in 2.1.

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carrying parts and the exposed nonmetal carrying parts. Evidence of

3.3.1 Body and cover. The body and cover of the dough trough shall be of panel construction having an inner liner of corrosion-resisting steel specified in 3.2.2, and insulation core specified in 3.2.9 and an outer liner of aluminum alloy specified in 3.2.3.1. The insulation shall be bonded to the panels with adhesive specified in 3.2.10. The corrosion-resisting steel, aluminum alloy sheet, and insulation prior to bonding, shall be thoroughly cleaned with a suitable solvent, such as toluene. All inside joints shall be filled with solder that is free of cadmium, antimony, bismuth, lead, and other toxic materials. Handles for the cover shall be as specified in 3.2.6. The body handles and tie bars shall be formed, welded, and attached to the body as shown on Drawing 5-11-303.

3.4 Finish.

3.4.1 Surface preparation. Prior to assembly, all aluminum alloy parts shall be given a treatment conforming to class 1A of MIL-C-5541. All surfaces of steel and aluminum alloy, except aluminum alloy rivets, which are in direct contact with each other, shall be covered with one coat of primer specified in 3.2.11.

3.4.2 Cleaning. After assembly of the trough, all exterior surfaces shall be thoroughly cleaned and degreased followed by a thorough drying. Special attention shall be given to drying out crevices and other places difficult to dry.

3.4.3 Primer. All exterior surfaces of the cover and body shall be coated with zinc chromate primer specified in 3.2.11.

3.4.4 Painting. When thoroughly dry, the primer shall be coated with enamel specified in 3.2.12. The coating shall level out to a smooth and uniform film without application marks, orange peel, runs, wrinkles, drops or areas of thin film or no film, and shall be free of grit or coarse particles and blisters. All surfaces of the trough and cover which will be in direct contact with the dough shall remain unfinished.

3.5 Marking for identification. The trough shall be marked by indenting of the letters "U.S.", the manufacturers' name, trade name or trademark and date of manufacture of the size, and in the location shown on Drawing 5-11-299.

3.6 Workmanship. Components shall be assembled in a neat manner. Painted areas shall be smooth, uniformly coated, free of runs, wrinkles, orange peel, grit, or areas of no film.

3.6.1 Metal fabrication. All metal used in the fabrication of the dough trough shall be free of kinks, sharp bends, and slivers. Shearing shall be done neatly and accurately. Bends shall be made accurately to provide uniformity of size and shape. All pins, studs, and screw holes shall be accurately drilled and all burrs removed.

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3.6.2 Riveting. Holes for rivets shall be accurately punched or drilled. Rivets shall be tight, fully peened, and heads in full contract with the surface of the metal. All rivets where possible shall be driven from inside of the dough trough and peened on the outside. Stems of all exposed rivets shall be ground smooth.

3.6.3 Soldering. All inside seams shall be soldered and have a fillet of not less than 1/4 inch radius and shall be finished smooth with no holes, fissures, fins, or rough spots.

3.6.4 Welding. Surfaces of all parts to be welded shall have oxide, scale, paint, grease and other foreign matter removed. The welds shall be continuous, adherent, smooth and free of cracks, pits, imbedded foreign material, flux and scale.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for defects specified in 4.4.3.1. The presence of any defect or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.1.1 Insulation, rubber cellular. The insulation shall be tested for water absorption, density, and compression strength. Testing shall be performed on the insulation material in accordance with the following:

Characteristic	Test methods
Water absorption	FED-STD-601, Method No. 12411
Density	ASTM D 1565-70
Compressive strength	ASTM D 1621-64

Three sheets of any size insulation shall be selected at random from the amount of insulation to be used under any one contract. The sample unit shall be a 12 inch by 12 inch piece of the insulation material. The inspection level shall be S-1 with an AQL, expressed in terms of defects per hundred units, shall be 2.5. All the test requirements shall be applicable to the sample unit.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination shall be made during the manufacturing process to insure that the operations listed below are performed:

(a) Aluminum surfaces are treated (see 3.4.1).
(b) Stainless steel sheets, aluminum sheet and insulation area cleaned prior to bonding (see 3.3.1).
(c) Aluminum alloy and steel surfaces which are in direct contact are coated with primer (see 3.4.1).
(d) Surfaces are cleaned prior to welding (see 3.6.4).
(e) Assembled dough trough is thoroughly cleaned, degreased and dried prior to application of finish (see 3.4.2).
(f) Primer is applied to the exterior surfaces prior to painting (see 3.4.3).

Whenever nonconformance is noted, correction or replacement shall be made to affected items and process.

4.4.2.2 In-process dimensional examination. Examination shall be made of all components and subassemblies for detailed dimensions shown on the drawings listed in section 2. The inspection lot shall be all like assemblies components and subassemblies offered for inspection at one time. The inspection level shall be level II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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4.4.2.3 In-process examination. The defects found during examination shall be classified as in table I. The sample unit shall consist of one side, end, bottom, left cover or right cover panel, prior to sealing of each panel. The inspection level shall be level II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Insulation	Omitted, not type specified	X	
	Not butted tight against panel frame tubing	X	
	Evidence that the insulation is not firmly cemented with specified adhesive to both panels and tubing	X	
	Not recessed in left and right cover panels to receive handle body		X
	Crumbled or deteriorated	X	
	Evidence that scraps or waste materials are being used or voids in panel	X	

4.4.3 End item inspection. A lot shall consist of all dough troughs offered for inspection at one time. The sample unit shall be one completely fabricated dough trough.

* 4.4.3.1 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expressed in units of dough troughs. The sample unit shall be one dough trough. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Finish not as specified	X	
	Not finished where required	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Finish (cont'd)			
Surface appearance of enameled parts (exterior of body, cover, and tie bar assembly)	Finish not continuous, smooth, uniform, or adherent (blisters, flaky, or tacky)	X	
	Thin film (primer shows through)		X
	Surface coarse, not free from grit, orange peel, runs, wrinkles, or drops		X
Construction and workmanship	Any part missing or in wrong location	X	
	Punctured, dented, buckled, deteriorated, or out of shape	X	
	Pronounced tool or forming marks		X
	Sharp burrs, sliver or splinters	X	
	Any part not securely attached	X	
Body, cover, and tie assembly	Handle does not lay flush with top of cover or is improperly installed	X	
	Tie bar assembly not properly installed or does not function properly	X	
	Tie bar assembly loose, or not riveted as specified	X	
Interior (body and cover)	Surface contains seams	X	
	Joints not completely closed with solder	X	
	Paint on interior surfaces	X	
	Soldered joints not finished smooth	X	
	Dirt or other foreign matter inside compartment		X
Welding	Missing, incomplete, burn holes, cracks, fractured, or not fused	X	
	Slag inclusion, undercut, not smooth and uniform	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding (cont'd)	Poor surface appearance of weld as from craters, piling of deposited metal, evidence that weld has not been ground or filled where required, or grinding or filling not neatly accomplished	X	
Solder	Missing, not adherent, or incomplete	X	
	Not clean (flux or flux residue not removed), surfaces not smooth, not neatly finished, or pinholes in solder	X	
Rivets	Missing or broken, not approved type	X	
	Rivets not ground flush where required	X	
	Rivets not inserted from the inside when attaching doubler plate to vertical closure		X
Identification marking	Missing, incomplete, not legible, not in proper location or incorrect		X

* 4.4.3.2 End item dimensional examination. The end item shall be examined for conformance to dimensions specified herein. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of dough troughs. The sample unit shall be one dough trough. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging inspection. Examination shall be made to determine that packaging requirements as specified in section 5 is in conformance with MIL-B-2316.

5. PACKAGING

5.1 Levels A, B, and C. Each dough trough shall be preserved, packaged, packed and marked for each level of protection in accordance with MIL-B-2316 (see 6.2).

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6. NOTES

6.1 Intended use. The dough trough is intended for the fermentation of dough for field ration bread. The dough trough is used with the monorail system of the Dough Mixing and Make Up Outfit, Trailer Mounted, covered by MIL-D-43490, which is a component of the Bakery Plant, Mobile.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.4).
c. Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

* 6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

* 6.4 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample and consist of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicated where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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Review activity:

Army - MD

Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-2222F	Dough Trough: Baker's (For Mobile Bakery M-534)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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